Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “free fall” and “initial” to explain your model or result. Sentence starters are optional.

free fall: motion with gravity as the only force

initial: at the beginning of the interval; subscript i

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-40, 60) km with bounds ±4 km in x and ±6 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

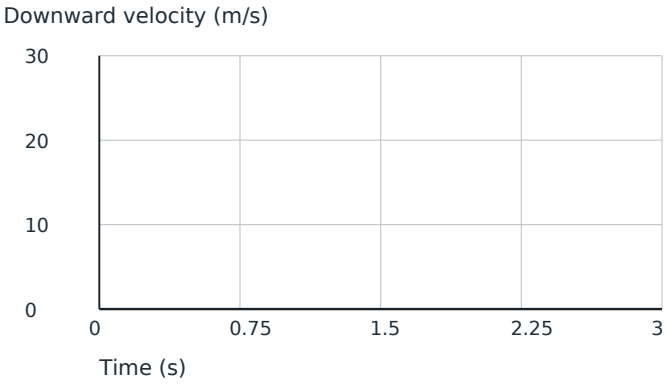
02 Worked example: explain the model

Reference: $v_f = v_i + g\,\Delta t$; $\Delta y = v_i\,\Delta t + \frac{1}{2}g(\Delta t)^2$. Downward positive for these equations; g = 10 m/s². Ideal free fall: gravity is the only force.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive.
Explain why the positions are not equally spaced.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two balls of different mass are released together in a vacuum from the same height.
Compare acceleration and arrival time.

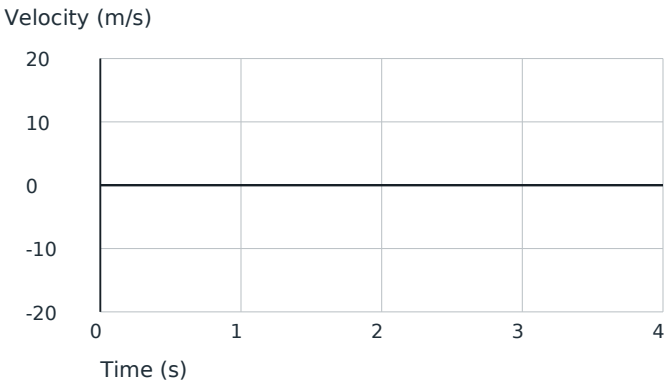
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Apply the physics

A ball is thrown straight upward at 20 m/s. Take upward as positive; use g = 10 m/s² and ignore air resistance. Complete its velocity-time graph. When does it reach the top, and what is its velocity at 3 s?

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An a-versus-F_{net} graph has slope 2 (m/s²)/N.
Find mass and acceleration at 3 N.
Explain why a driving-force plot can have an offset.
Use the table to keep the controlled and changed quantities distinct.

Word help: mass = a measure of how strongly an object resists acceleration

Sentence starter: Under the assumption …, the evidence … supports … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

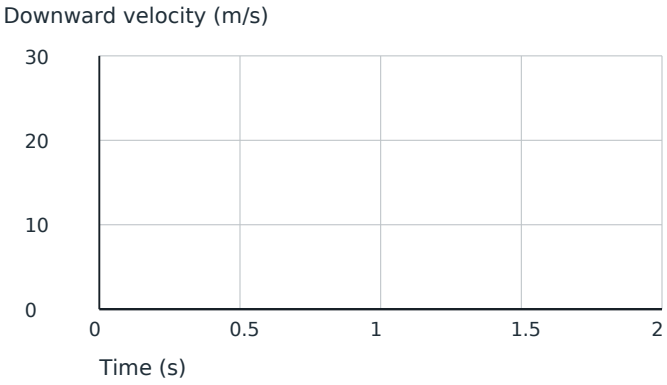
Sentence starters are optional. You may plan in any language.

07 Exit check

A ball starts downward at 5 m/s. It falls for 2 s with g = 10 m/s² and negligible air resistance.
Draw its velocity-time graph.
Use slope and area to find its final velocity and downward displacement.

Word help: negligible = small enough to leave out of this model

Sentence starter: I found … [unit]. This value means … in this situation.

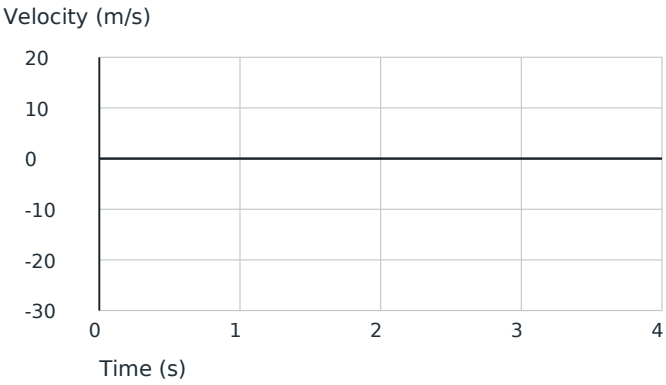


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A ball is thrown up at 15 m/s from a 20 m platform.
Use g=10 m/s².
Find its height at 2 s and its time to reach the ground.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3